

BBC micro:bit: Heart LED

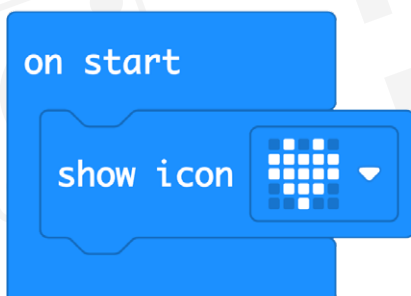
1. Getting Started

What you will need:

- micro:bit
- Microsoft MakeCode
- USB data cable, if using a desktop or laptop computer
- battery pack, if using a mobile device

2. Code

- Inside a **on start** block, slot in the **show icon Basic** block.
- Ensure the heart is showing in the drop-down menu.



3. Test

To test your code, select the play button to start the **simulator**.

4. Download

- Click on the download button to transfer the code to your micro:bit.
- Follow the directions on screen.
- When the code is transferred you should see a heart appear in the LEDs.

BBC micro:bit: Beating Heart

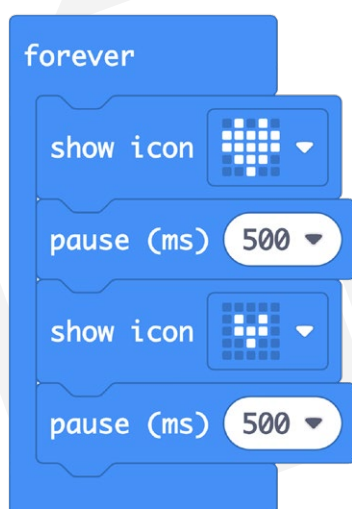
1. Getting Started

What you will need:

- micro:bit
- Microsoft MakeCode
- USB data cable, if using a desktop or laptop computer
- battery pack, if using a mobile device

2. Code

- Inside a **forever Basic** block, slot in the **show icon Basic** block.
- Ensure the heart is showing in the drop-down menu.
- Add a **pause (ms)** block and select **500** from the drop-down menu.
- Add another **show icon Basic** block.
- Select the smaller heart from the drop-down menu.
- Add another **pause (ms)** block and select **500** from the drop-down menu.



3. Test

To test your code, select the play button to start the **simulator**.

4. Download

- Click on the download button to transfer the code to your micro:bit.
- Follow the directions on screen.
- When the code is transferred you should see a beating heart appear in LEDs.

Challenge

Can you alter the code so that the heart only beats when **button A** is pressed?

BBC micro:bit: Name Badge

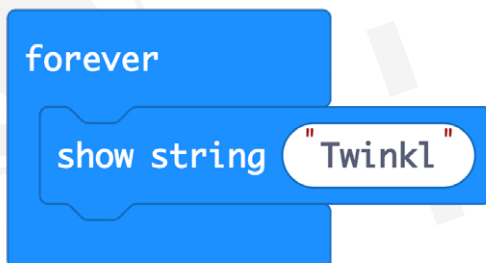
1. Getting Started

What you will need:

- micro:bit
- Microsoft MakeCode
- USB data cable, if using a desktop or laptop computer
- battery pack, if using a mobile device

2. Code

- Select the **on Input** block and select **shake** from the drop-down menu.
- Insert a **forever** block and type your name in the white space.



3. Test

To test your code, select the play button to start the **simulator**.

4. Download

- Click on the download button to transfer the code to your micro:bit.
- Follow the directions on screen.

Challenge

Can you alter the code so that your name only shows when you shake the micro:bit?

1. Getting Started

What you will need:

- micro:bit
- Microsoft MakeCode
- USB data cable, if using a desktop or laptop computer
- battery pack, if using a mobile device

2. Code

- Select the **on button pressed Input** block and select **A** from the drop-down menu.
- Insert a **show number Basic** block.
- Find the **temperature (°C) Input** block and move this into the white space on the show number **Basic** block.



3. Test

To test your code, select the play button to start the **simulator**.

4. Download

- Click on the download button to transfer the code to your micro:bit.
- Follow the directions on screen.

BBC micro:bit: Rock, Paper, Scissors

1. Getting Started

What you will need:

- micro:bit
- Microsoft MakeCode
- USB data cable, if using a desktop or laptop computer
- battery pack, if using a mobile device

2. Code

- Select the **on shake Input** block.
- Go into the **Variables** category and make a new variable with the name 'tool'.
- Add a **set tool to Variables** block.
- Find the **pick random 0 to 10 Math** block, and change the 10 to a **2**.
- Insert the **pick random 0 to 2 Math** block over the 0 in the **set tool to Variables** block.
- Insert an **if...then...else Logic** block.
- Find the **0 = 0 Logic** block and slot this over **true** on the **if...then...else** block.
- Select the **tool Variables** block and slot this over the first **0** in the **0 = 0** block.
- Click the **plus** symbol below the **else** in the **if...then...else Logic** block.
- Right-click the **tool = 0** block and select **Duplicate** and change the 0 to be a **1**.
- Slot the duplicated block over **true** on the second space in the **if...then...else** block.
- Select three **show icon Basic** blocks and place these in the **if...then...else** block.
- Change the icons in the drop-down menus to represent a rock, paper and scissors.

3. Test

To test your code, select the play button to start the **simulator**.

4. Download

- Click on the download button to transfer the code to your micro:bit.
- Follow the directions on screen.

